Ahmedabad University

CSE623 Machine Learning

Weekly Report 3

Section 1

Submitted to Faculty: Prof. Mehul Raval

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1. Tasks completed this week

Last week, our team trained a Random forest classifier, which is an ensemble model which constructs multiple decision trees; each of which takes a random subset of features and makes a prediction. The majority of which is taken as the final prediction. The model achieved an overall test accuracy of 100%, with a mean accuracy of 1.0 across all classes. Additionally, precision and recall scores for each class—Hard stop, Momentary stop, and None—were all equal to 1.0. However, our team believed the perfect performance of the model is rather misleading, especially since most of the time, such predictions are often the result of model overfitting. Now, our goal this week is to understand what is causing the model to give this kind of prediction. Is the issue due to the model or the dataset?

To answer this question, the team decided to implement an unsupervised model to determine whether the three classes of stops were natural clusters in the feature space, with no labels being used whatever. In this regard, use of K-Means clustering was made. K-Means is based on the idea of minimising the distance between the data point and the nearest cluster centre to group the data points into k clusters until the clusters become stable. Here it was the simplest option since the number of clusters was known- there were three classes- k=3.

Tasks

- Implementation of an Unsupervised Clustering Model (Kmeans).
- Analysis and Comparison of results with that of Random Forest Classifier.
- Performing Data Analysis to understand the limitations of the dataset.

2. Results and Limitations:

After implementing the Kmeans Clustering model, we used the Principal component analysis to project our clusters into the 2D PCA space in order to visualize them. The following is the result:

This image shows that the blue spots are undergoing no stopping behaviour, whereas the purple ones are showing the momentary stop and the yellow is the one with a hard stop, as there are many hard stops, but all have the same velocity, so they are piled up, one on another, which we can view in 3D plot.

There was a problem with the dataset itself, which is why the result is as it is.

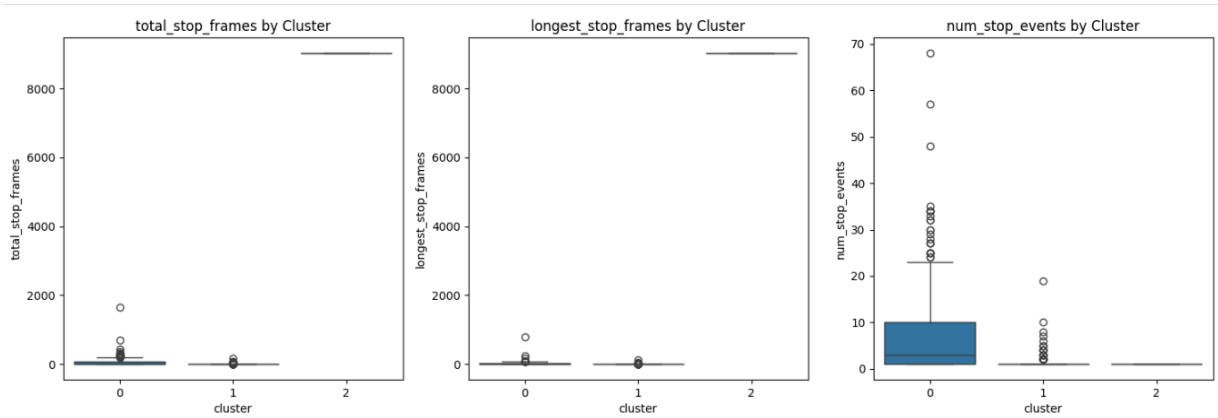
Certainly, the above figure showcases that the model seems to be overfitting due to the dataset. But before declaring that the model is memorizing the patterns, our team decided to test the model with unseen data by providing the model with a new set of dataset. As it can be seen below, the first two points which tell the stop duration clearly indicate that this is a datapoint of a moving object that moves for a while and then stops for a prolonged period of time. This tells us that this should be classified under Hard stop or [2] cluster. But when we gave the model this to predict, its prediction was [0] or momentary stop. It is clear by this that the model is memorizing the patterns.

```
[29]: new_data = [[5000, 5000, 12, 210, 4.0, 12.0, 2.8, 2.0, -2.5, 1.5, 20.0,300]]
      new_scaled = scaler.transform(new_data)
      cluster = kmeans.predict(new_scaled)
      print(cluster)

      [0]
```

The sample size for the hard stop class is relatively small compared to the other classes, resulting in a noticeable class imbalance. Additionally, because the trajectories of hard stop objects were derived from first-frame values due to insufficient data, the dataset lacks the natural variability typically observed in object clustering. As a consequence of these limitations, the model appears to be memorizing specific patterns rather than learning generalized representations.

The lack of variance in hard stop data can be observed if we boxplot the dataset feature by feature for each cluster.



Aspect	Random Forest Classifier	K-Means Clustering
Labels	Yes	No
Goal	Classify each vehicle's stop type	Discover natural groupings in data
Result	100% accuracy, F1 = 1.0 for all classes	3 clusters matching the 3 stop classes

Comparison of Random Forest Classifier and K-Means Clustering

3. Agenda for Next Week

The tasks planned to be done next week are the following:

- Checking whether the model truly generalizes or is overfitting the dataset-specific hard stop profile by evaluating on augmented samples or synthetic hard stop samples using more varied feature values.
- Compare the performance of other models, such as XGBoost (supervised classification) and DBSCAN (unsupervised clustering), with that of the Random Forest and K-Means.
- Continue to increase the size of the hard stop data with additional examples whose trajectory profiles naturally vary, to decrease class imbalance and add natural feature variance to the hard stop class.
- Test edge cases, e.g. the problem of partial stops and near-zero velocity, to test the capability of the current features and thresholds to respond to ambiguous stopping behaviour.